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Trap device and substrate processing apparatus comprising the same

Abstract

The present disclosure provides a substrate processing apparatus capable of stably controlling a meniscus position. The substrate processing apparatus comprises a head unit configured to discharge a medical fluid; a reservoir configured to store the medical fluid and supply the medical fluid to the head unit; a pressure control unit configured to control pressure in the reservoir; and a trap unit disposed between the reservoir and the pressure control unit and configured to trap a mist generated by the reservoir.

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Background/Summary**CROSS-REFERENCE TO RELATED APPLICATION**

(1) This application claims priority from Korean Patent Application No. 10-2021-0147339 filed on Oct. 29, 2021 in the Korean Intellectual Property Office, and all the benefits accruing therefrom under 35 U.S.C. 119, the contents of which in its entirety are herein incorporated by reference.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a trap device and a substrate processing apparatus comprising the same.

2. Description of the Related Art

(3) In order to manufacture display devices such as an LCD panel, a PDP panel and an LED panel, printing is performed on a substrate using an inkjet head. A meniscus position in a nozzle of the inkjet head is one of the central elements that determine injection characteristics of ink. The meniscus position may be controlled by a meniscus pressure controller (MPC).

SUMMARY

(4) Aspects of the present disclosure provide a substrate processing apparatus capable of stably controlling a meniscus position without malfunction.

(5) Aspects of the present disclosure also provide a trap device used in a substrate processing apparatus.

(6) The technical aspects of the present disclosure are not restricted to those set forth herein, and other unmentioned technical aspects will be clearly understood by one of ordinary skill in the art to which the present disclosure pertains by referencing the detailed description of the present disclosure given below.

TECHNICAL SOLUTION

(7) According to an aspect of the present disclosure, there is provided a substrate processing apparatus, comprising: a head unit configured to discharge a medical fluid; a reservoir configured to store the medical fluid and supply the medical fluid to the head unit; a pressure control unit configured to control pressure in the reservoir; and a trap unit disposed between the reservoir and the pressure control unit and configured to trap a mist generated by the reservoir.

(8) According to another aspect of the present disclosure, there is provided a substrate processing apparatus, comprising: a stage configured to process a substrate; a gantry disposed to cross the stage; and an inkjet head module installed in the gantry and configured to discharge ink to the substrate, wherein the inkjet head module comprises: a head unit configured to discharge the ink; a reservoir configured to store the ink and supply the ink to the head unit; a pressure control unit configured to control pressure in the reservoir; and a trap unit disposed between the reservoir and the pressure control unit and configured to trap a mist generated by the reservoir, wherein the trap unit comprises: a body; a first line installed in the body, connected to the reservoir and extending in a first direction; a second line installed in the body, connected to the pressure control unit and extending in a second direction; a third line installed to penetrate the first line and the second line from an upper surface of the body; and a trap layer installed on an inner wall of the first line or an inner wall of the second line and configured to trap the mist.

(9) According to an aspect of the present disclosure, there is also provided a trap device comprising: a body; a first line installed in the body, connected to an inlet and extending in one direction; a second line installed in the body, connected to an outlet and extending in the one direction; a third line installed to penetrate the first line and the second line from an upper surface of the body; and a trap layer installed on an inner wall of the first line or an inner wall of the second line and configured to trap a mist.

(10) Specific details of other embodiments are included in the detailed description and the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other aspects and features of the present disclosure will become more apparent by describing in detail exemplary embodiments thereof with reference to the attached drawings, in

which:

- (2) FIG. 1 is a block diagram illustrating a substrate processing apparatus according to some embodiments of the present disclosure;
- (3) FIG. 2 is a view illustrating in detail the supply reservoir, the trap unit, and the pressure control unit of FIG. 1.
- (4) FIG. 3 is a cross-sectional view illustrating a trap unit according to a first embodiment of the present disclosure;
- (5) FIG. 4 is a diagram illustrating a mist motion in the trap unit of FIG. 3;
- (6) FIG. 5 is a cross-sectional view illustrating the trap unit according to a second embodiment of the present disclosure;
- (7) FIG. 6 is a cross-sectional view illustrating the trap unit according to a third embodiment of the present disclosure;
- (8) FIG. 7 is a cross-sectional view illustrating the trap unit according to a fourth embodiment of the present disclosure;
- (9) FIG. 8 is a cross-sectional view illustrating a trap unit according to a fifth embodiment of the present disclosure;
- (10) FIG. 9 is an exemplary diagram illustrating a facility to which the substrate processing apparatus according to some embodiments of the present disclosure is applied; and
- (11) FIG. 10 is a flowchart illustrating the substrate processing method according to some embodiments of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(12) Hereinafter, embodiments of the present disclosure will be described with reference to the attached drawings. The merits and characteristics of the present disclosure and a method for achieving the merits and characteristics will become more apparent from the embodiments described in detail in conjunction with the accompanying drawings. However, the present disclosure is not limited to the disclosed embodiments, but may be implemented in various different ways. The embodiments are provided to only complete the disclosure of the present disclosure and to allow those skilled in the art to understand the category of the present disclosure. The present disclosure is defined by the category of the claims. Like numbers refer to like elements throughout the description of the figures.

(13) The spatially relative terms “below,” “beneath,” “lower,” “above,” “upper” may be used to easily describe the correlation of a device or components with other devices or components. Spatially relative terms are to be understood as including terms in different directions of the device in use or operation in addition to the directions shown in the figures. For example, when flipping a device shown in the figure, a device described as “below” or “beneath” of another device may be placed “above” of another device. Thus, the exemplary term “below” can encompass both an orientation of above and below. The device can also be oriented in other directions, so that spatially relative terms can be interpreted according to orientation.

(14) Although the first, second, etc. are used to describe various elements, components and/or sections, these elements, components and/or sections are of course not limited by these terms. These terms are only used to distinguish one element, component or section from another element, component or section. Therefore, the first device, the first component, or the first section mentioned below may be a second device, a second component, or a second section within the technical spirit of the present disclosure.

(15) Hereinafter, preferred embodiments according to the present disclosure are described in detail with reference to the accompanying drawings. The same or similar elements are assigned the same reference numerals irrespective of their reference numerals, and a redundant description thereof is omitted.

(16) FIG. 1 is a block diagram illustrating a substrate processing apparatus according to some embodiments of the present disclosure. FIG. 2 is a view illustrating in detail the supply reservoir,

the trap unit, and the pressure control unit of FIG. 1.

(17) First, referring to FIG. 1, a substrate processing apparatus **1** according to some embodiments of the present disclosure includes a head unit **210**, a supply reservoir **220**, a pressure control unit **230**, a buffer reservoir **240**, and a trap unit **100**.

(18) The head unit **210** receives a medical fluid from the supply reservoir **220** and discharges the received medical fluid onto the substrate. The head unit **210** includes a plurality of nozzles configured to discharge the medical fluid onto the substrate. The medical fluid supplied to the head unit **210** may be supplied by gravity from the supply reservoir **220** disposed in an upper portion of the head unit **210**, but the present disclosure is not limited thereto.

(19) The supply reservoir **220** is provided in the upper portion of the head unit **210**. The supply reservoir **220** receives and stores the medical fluid from the buffer reservoir **240** disposed in an upper portion of the supply reservoir **220**.

(20) The pressure control unit **230** is connected to the supply reservoir **220** and controls pressure in the supply reservoir **220**. The pressure control unit **230** may be a meniscus pressure controller (MPC).

(21) The pressure control unit **230** supplies positive pressure and/or negative pressure into the supply reservoir **220** to control the pressure in the supply reservoir **220**.

(22) Specifically, the pressure control unit **230** may control the supply of the medical fluid from the supply reservoir **220** to the head unit **210** by controlling the pressure in the supply reservoir **220**. For example, the supply of the medical fluid from the supply reservoir **220** to the head unit **210** is performed by gravity, and the suspension of the medical fluid supply is performed by allowing the pressure control unit **230** to supply the negative pressure into the supply reservoir **220**.

(23) Furthermore, the pressure control unit **230** controls the medical fluid to be in a concave meniscus state (i.e., a state where a surface of the medical fluid increases in the center relative to the surroundings due to a surface tension) in a plurality of nozzle ends provided in the head unit **210**. When the medical fluid is in the concave meniscus state, the medical fluid does not flow down from the end of the nozzle, which can reduce substrate defects.

(24) Meanwhile, the buffer reservoir **240** receives and stores the medical fluid from an external medical fluid supply source (not shown), and supplies the received medical fluid to the supply reservoir **220**. When the medical fluid is supplied from the medical fluid supply source (not shown) to the buffer reservoir **240**, the inside of the medical fluid supply source is pressurized. When the medical fluid supply source is directly connected to the supply reservoir **220**, the internal pressure of the supply reservoir **220** is affected by the medical fluid supply source. However, since there is the buffer reservoir **240** between the medical fluid supply source and the supply reservoir **220**, the influence of the medical fluid supply source on the supply reservoir **220** may be blocked.

(25) The trap unit **100** is installed between the supply reservoir **220** and the pressure control unit **230**. The trap unit **100** traps a mist generated in the supply reservoir **220**.

(26) Herein, referring to FIG. 2, a heater **222** is installed in the supply reservoir **220** so as to control the temperature of the medical fluid. Although FIG. 2 illustrates that the heater **222** is installed on the bottom surface of the supply reservoir **220**, the present disclosure is not limited thereto. In other words, the heater **222** may also be installed on a sidewall or an upper surface of the supply reservoir **220**.

(27) The pressure control unit **230** is connected to the supply reservoir **220** through tubes **229** and **239**. An interlock sensor **232** is installed in the tube **239**. The interlock sensor **232** may be, for example, a liquid detection sensor. The pressure control unit **230** is vulnerable to liquid (i.e., medical fluid, etc.). When the pressure control unit **230** is exposed to liquid, malfunction may occur or meniscus control may become inaccurate. Accordingly, when the liquid is detected in the interlock sensor **232** before the liquid reaches the pressure control unit **230**, an interlock for stopping the operation occurs.

(28) Meanwhile, the trap unit **100** is disposed between the supply reservoir **220** and the pressure

control unit **230**. In other words, the trap unit **100** is connected to the supply reservoir **220** through the tube **229** and is connected to the pressure control unit **230** through the tube **239**.

(29) As described above, since the heater **222** is installed in the supply reservoir **220**, mist (i.e., medical fluid mist or vaporized medical fluid) may be generated by the heating operation of the heater **222**. When the mist is delivered to the pressure control unit **230**, this may cause malfunction of the pressure control unit **230**. Alternatively, the installation purpose of the interlock sensor **232** is to detect the occurrence of overflow; however, when the mist is detected by the interlock sensor **232**, the interlock may take place. Although the medical fluid has not overflowed from the supply reservoir **220**, the interlock sensor **232** may recognize that the medical fluid has overflowed (i.e., a malfunction of the interlock sensor **232** occurred).

(30) Accordingly, the trap unit **100** is disposed below the interlock sensor **232** to block the mist so that the mist generated by the supply reservoir **220** is not delivered to the interlock sensor **232** or the pressure control unit **230**.

(31) The trap unit **100** is inclined at an acute angle (θ) with respect to an upper surface **220a** of the supply reservoir **220**. As will be described below, a contact surface between the mist and the trap unit **100** is increased so that the mist may be more trapped in the trap unit **100**. Furthermore, when the liquefied medical fluid from the trap unit **100** drops to the supply reservoir **220**, it flows slowly along the inclined tube **229** to prevent bubbles from occurring in the supply reservoir **220**.

(32) Hereinafter, various embodiments of the trap unit **100** will be described with reference to FIGS. 3 to 7.

(33) FIG. 3 is a cross-sectional view illustrating a trap unit according to a first embodiment of the present disclosure. FIG. 4 is a diagram illustrating a mist motion in the trap unit of FIG. 3.

(34) First, referring to FIG. 3, the trap unit **100** according to a first embodiment of the present disclosure includes a body **105**, a first line **110**, a second line **120**, a third line **130**, a first trap layer **116**, and a second trap layer **126**.

(35) The body **105** may be made of, for example, metal, and the first line **110**, the second line **120**, and the third line **130** are installed in the body **105**.

(36) The first line **110** is connected to the supply reservoir **220** (see FIG. 2) and extends in a first direction. The first trap layer **116** configured to trap the mist is installed on an inner wall of the first line **110**. The first trap layer **116** may include, for example, at least one of a mesh structure, a membrane structure and a filter structure, but the present disclosure is not limited thereto. The mesh structure may be made of metal such as SUS. The membrane structure and the filter structure may contain mist including pores inside, but the present disclosure is not limited thereto.

(37) The second line **120** is connected to the pressure control unit **230** and extends in a second direction. As illustrated in the drawings, the first direction and the second direction may be substantially the same direction (i.e., a parallel direction), but the present disclosure is not limited thereto. The second trap layer **126** configured to trap mist is installed on an inner wall of the second line **120**. The second trap layer **126** may include, for example, at least one of the mesh structure, the membrane structure and the filter structure, but the present disclosure is not limited thereto.

(38) The third line **130** connects the first line **110** to the second line **120**. For example, the third line **130** may be installed to penetrate the first line **110** and the second line **120** from an upper surface of the body **105**. A cover **140** may be installed on the upper surface of the body **105** to open or seal one side of the third line **130**. The cover **140** may be, for example, a screw, but the present disclosure is not limited thereto.

(39) The first line **110** includes a first connector **111** connected to the third line **130**. The first line **110** includes an inlet **118** disposed in a first side (e.g., the left side) centered on the first connection port **111** and a first buffer region **119** disposed in a second side (e.g., the right side) centered thereon. A first connection member **110a** for connection with the tube **229** is installed in the inlet **118**. The first connection member **110a** may be inserted/fixed in the body **105**.

(40) Similarly, the second line **120** includes a second connector **121** connected to the third line **130**.

The second line **120** includes an outlet **128** disposed in the second side (e.g., the right side) centered on the second connection port **121** and a second buffer region **129** disposed in the first side (e.g., the left side) centered thereon. A second connection member **120a** for connection with the tube **239** is installed in the outlet **128**. The second connection member **120a** may be inserted/fixed in the body **105**.

(41) In this way, the first line **110**, the second line **120**, and the third line **130** are configured so as to make the movement path of the mist as long as possible. When the movement path of the mist is lengthened, the mist can be liquefied in the trap unit **100** while the mist moves along the movement path.

(42) Herein, the mist motion in the trap unit **100** will be described with reference to FIG. **4**. For example, the first trap layer **116** and the second trap layer **126** are mesh structures made of metal SUS.

(43) The mist generated from the supply reservoir **220** is inserted into the trap unit **100** via the first connection member **110a** and the inlet **118** (see reference mark G1).

(44) A part of the mist inserted into the trap unit **100** increases to the first buffer region **119** and is trapped in the first buffer region **119** (see reference mark G2). The mist trapped in the first buffer region **119** is liquefied while it is in contact with the first trap layer **116** with a relatively low temperature, and the liquefied medical fluid flows downwards along the extending direction of the first line **110** and then comes out of the inlet **118**.

(45) In addition, a part of the mist inserted into the trap unit **100** reaches the third line **130** (see reference mark G3). To ensure that the mist can move along the third line **130**, the mist needs to pass through the first trap layer **116** with a relatively low temperature. In the process of passing through the first trap layer **116**, the temperature of the mist may decrease or the liquefaction of the mist may occur.

(46) In addition, a part of the mist inserted into the third line **130** reaches the second line **120** (see reference marks G4 and G5). To ensure that the mist can move along the second line **120**, the mist needs to pass through the second trap layer **126** with a relatively low temperature. In the process of passing through the second trap layer **126**, the temperature of the mist may decrease or the liquefaction of the mist may occur.

(47) A part of the mist reaching the second line **120** goes to the second buffer region **129** and is trapped in the second buffer region **129** (see reference mark G5). The mist trapped in the second buffer region **129** may be liquefied while it is in contact with the second trap layer **126** with a relatively low temperature, and the liquefied medical fluid may be collected in the second buffer region **129** or flow through the third line **130**.

(48) A part of the mist reaching the second line **120** may move in the direction of the outlet **128** (see reference mark G4). However, the mist is continuously in contact with the trap layers **116** and **126** with a relatively low temperature as it passes through a long movement path. Therefore, the mist does not reach the outlet **128** and is mostly liquefied.

(49) Meanwhile, the liquefied medical fluid remains in the trap unit **100** or is moved to the supply reservoir **220** through the tube **229**.

(50) When the cover **140** is opened, one side of the third line **130** extending to the upper surface of the body **105** is opened. Accordingly, the medical fluid remaining in the trap unit **100** may be extracted through the third line **130** extending to the upper surface of the body **105**.

(51) In addition, the trap unit **100** is inclined at an acute angle (θ) with respect to the upper surface **220a** of the supply reservoir **220**. Accordingly, the tube **229** configured to connect the trap unit **100** and the supply reservoir **220** is also inclined. As a result, the liquefied medical fluid flows slowly along the inclined tube **229** and drops into the supply reservoir **220**. Therefore, bubbles do not occur in the supply reservoir **220**.

(52) Accordingly, the trap unit **100** blocks the mist so that the mist generated in the supply reservoir **220** is not delivered to the interlock sensor **232** or the pressure control unit **230**. Therefore, it is

possible to prevent malfunction of the interlock sensor **232**, and to enable the pressure control unit **230** to stably operate.

(53) FIG. **5** is a cross-sectional view illustrating the trap unit according to a second embodiment of the present disclosure. Hereinafter, the differences from those described with reference to FIGS. **3** and **4** will be mainly described.

(54) Referring to FIG. **5**, in the trap unit **101** according to the second embodiment of the present disclosure, the first trap layer **116** may not be installed on the entire inner wall of the first line **110**, but may instead be installed only on a part of the inner wall of the first line **110**. The second trap layer **126** may not be installed on the entire inner wall of the second line **120**, but may instead be installed only on a part of the inner wall of the second line **120**.

(55) Although FIG. **5** illustrates that the first trap layer **116** is installed around the inlet **118** and is not installed in the first buffer region **119**, the present disclosure is not limited thereto. For example, the first trap layer **116** may not be installed around the inlet **118**, but may be installed in the first buffer region **119**.

(56) Although FIG. **5** illustrates that the second trap layer **126** is installed around the outlet **128** and is not installed in the second buffer region **129**, the present disclosure is not limited thereto. For example, the second trap layer **126** may not be installed around the outlet **128**, but may be installed in the second buffer region **129**.

(57) Even if the trap layers **116** and **126** are installed only on a part of the first line **110** and the second line **120**, the mist may be liquefied as it passes through a long path.

(58) FIG. **6** is a cross-sectional view illustrating the trap unit according to a third embodiment of the present disclosure. FIG. **7** is a cross-sectional view illustrating the trap unit according to a fourth embodiment of the present disclosure. Hereinafter, the differences from those described with reference to FIGS. **3** to **5** will be mainly described.

(59) Referring to FIG. **6**, unlike the trap unit **100** according to the first embodiment, the trap unit **102** according to the third embodiment of the present disclosure may not be provided with the first buffer region (see **119** of FIG. **3**). Referring to FIG. **7**, unlike the trap unit **100** according to the first embodiment, the trap unit **103** according to the fourth embodiment of the present disclosure may not be provided with the second buffer region (see **129** of FIG. **3**).

(60) FIG. **8** is a cross-sectional view illustrating a trap unit according to a fifth embodiment of the present disclosure. Hereinafter, the differences from those described with reference to FIGS. **3** and **4** will be mainly described.

(61) Referring to FIG. **8**, third lines **131**, **132** and **135** of a trap unit **104** according to the fifth embodiment of the present disclosure have a path longer than that of the third line **130** illustrated in FIG. **3**.

(62) For example, the first line **110** and the second line **120** may be parallel to each other, and the third line **131**, **132** and **135** may include an intermediate line **135**, a first vertical line **131**, and a second vertical line **132**. The intermediate line **135** is disposed between the first line **110** and the second line **120** and is parallel to the first line **110** and the second line **120**. The first vertical line **131** connects the intermediate line **135** to the first line **110**. The second vertical line **132** connects the intermediate line **135** to the second line **120**.

(63) The mist inserted into the inlet **118** may reach the outlet **128** only when it passes through the first line **110**, the first vertical line **131**, the intermediate line **135**, the second vertical line **132**, and the second line **120**. In other words, since an internal path of the trap unit **100** is significantly longer, the mist is liquefied as it passes through the internal path and does not reach the interlock sensor **232** or the pressure control unit **230**.

(64) FIG. **9** is an exemplary diagram illustrating a facility to which the substrate processing apparatus according to some embodiments of the present disclosure is applied.

(65) Referring to FIG. **9**, the facility includes a stage PT, a gantry **410**, an inkjet head module **420**, and a control unit **450**.

(66) The stage PT may extend long in a first direction Y, and the stage PT may move a substrate G in the first direction Y (see reference mark S). For example, a plurality of holes are formed in the stage PT, and gas may come out through the holes to levitate a manufacturing substrate. In a state where the manufacturing substrate is levitated, a holder may hold and move the substrate, but the present disclosure is not limited thereto.

(67) The gantry **410** is disposed on the stage PT. The gantry **410** is disposed to cross the stage PT. The gantry **410** is disposed to extend in a second direction X different from the first direction Y.

(68) The inkjet head module **420** may be installed in the gantry **410** and may move in the second direction X along the gantry **410** (see reference numeral W). The inkjet head module **420** corresponds to the apparatus **1** described with reference to FIGS. **1** to **8**.

(69) In other words, the inkjet head module **420** includes the head unit configured to discharge the ink, the reservoir configured to store the ink and supply the medical fluid to the head unit, the pressure control unit configured to control the pressure in the reservoir, and the trap unit disposed between the reservoir and the pressure control unit to trap the mist generated in the reservoir.

(70) Herein, the trap unit may include the body, the first line installed in the body, connected to the reservoir and extending in the first direction, the second line installed in the body, connected to the pressure control unit and extending in the second direction, the third line installed to penetrate the first line and the second line from the upper surface of the body, and a trap layer installed on the inner wall of the first line or the inner wall of the second line and configured to trap the mist. Herein, the body is inclined at the acute angle with respect to the upper surface of the reservoir.

(71) The first line includes the first connector connected to the third line, and the first line further includes the inlet disposed in the first side centered on the first connector, and the first buffer region disposed in the second side centered thereon. The second line includes the second connector connected to the third line, and the second line further includes the outlet disposed in the second side centered on the second connector, and the second buffer region disposed in the first side centered thereon.

(72) In addition, while the substrate G on the stage PT moves in the first direction Y (i.e., a swathing operation), the inkjet head module **420** may discharge droplets onto the substrate G while moving in the second direction X.

(73) The control unit **450** controls the stage PT, the inkjet head module **420** and so forth. The control unit **450** is connected to a memory (not shown), and instructions for operating the stage PT, the inkjet head module **420** and so forth are stored in the memory.

(74) FIG. **10** is a flowchart illustrating the substrate processing method according to some embodiments of the present disclosure.

(75) Referring to FIGS. **1** to **3** and **10**, the substrate processing apparatus **1** including the head unit **210**, the supply reservoir **220**, the pressure control unit **230**, and the trap unit **100**.

(76) Furthermore, the temperature of the medical fluid in the supply reservoir **220** is controlled using the heater **222** installed in the supply reservoir **220** (S401).

(77) The mist (i.e., medical fluid mist or vaporized medical fluid) is generated by the heating operation of the heater **222** (S403).

(78) Furthermore, the mist generated in the supply reservoir **220** is delivered to the trap unit **100** through the tube **229**. The mist is liquefied while moving along the path in the trap unit **100**. In other words, the mist may be liquefied by the trap layers **116** and **126** and/or the long movement path (i.e., the first line **110**, the second line **120** or the third line **130**). The liquefied medical fluid returns to the supply reservoir **220** along the tube **229** or is trapped in the trap unit **100** (S405).

(79) Then, the medical fluid remaining in the trap unit **100** is removed (S407). When the cover **140** that seals one side of the third line **130** is opened, one side of the third line **130** extending to the upper surface of the body **105** is opened. Accordingly, the medical fluid remaining in the trap unit **100** may be extracted via the third line **130** extending to the upper surface of the body **105**.

(80) Although the embodiments of the present disclosure have been described above with reference

to the accompanying drawings, the present disclosure is not limited to the disclosed embodiments, but may be implemented in various different ways, and the present disclosure may be embodied in many different forms without changing technical subject matters and essential features as will be understood by those skilled in the art. Therefore, embodiments set forth herein are exemplary only and not to be construed as a limitation.

Claims

1. A substrate processing apparatus, comprising: a head unit configured to discharge a medical fluid; a reservoir configured to store the medical fluid and supply the medical fluid to the head unit; a pressure control unit configured to control pressure in the reservoir; and a trap unit disposed between the reservoir and the pressure control unit and configured to trap a mist generated by the reservoir, wherein the trap unit comprises: a body, a first line installed in the body, connected to the reservoir and extending in a first direction, a second line installed in the body, connected to the pressure control unit and extending in a second direction, a third line installed from an upper surface of the body to penetrate the first line and the second line, a trap layer configured to trap the mist is installed on an inner wall of the first line or an inner wall of the second line, and a cover installed on the upper surface of the body to open or seal one side of the third line.
 2. The substrate processing apparatus of claim 1, wherein the trap layer includes at least one of a mesh structure, a membrane structure and a filter structure.
 3. The substrate processing apparatus of claim 1, wherein the body is inclined at an acute angle with respect to an upper surface of the reservoir.
 4. The substrate processing apparatus of claim 1, wherein the first line includes a first connector connected to the third line, and the first line further includes an inlet disposed in a first side centered on the first connector and a first buffer region disposed in a second side centered on the first connector.
 5. The substrate processing apparatus of claim 1, wherein the second line includes a second connector connected to the third line, and the second line further includes an outlet disposed in a second side centered on the second connector and a second buffer region disposed in a first side centered on the second connector.
 6. The substrate processing apparatus of claim 1, wherein the first line and the second line are parallel to each other, and the third line comprises: an intermediate line disposed between the first line and the second line and parallel to the first line and the second line; a first vertical line configured to connect the intermediate line and the first line; and a second vertical line configured to connect the intermediate line and the second line.
 7. The substrate processing apparatus of claim 1, wherein a heater configured to control the temperature of the medical fluid is installed in the reservoir.
 8. The substrate processing apparatus of claim 1, wherein an interlock sensor is installed in a tube configured to connect the trap unit and the pressure control unit.
 9. The substrate processing apparatus of claim 1, wherein the trap unit is positioned below an interlock sensor configured to detect liquid.
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